COFFEE SHOP SALES ANALYSIS – QUERY REPORT

I KPI's

a) Total Revenue

```
select
        sum(transaction_qty*unit_price) as total_revenue
from coffee_shop_sales;
```

	total_revenue
▶	698812.33

b) Total item quantity sold

```
select
        sum(transaction_qty) as total_quantity
from coffee_shop_sales;
```

total_quantity
214470

c) Total Orders

```
select
        count(transaction_id) as total_orders
from coffee_shop_sales;
```

total_orders
149116

d) Average Order Value (AOV)

```
select
 sum(transaction_qty*unit_price)/count(transaction_id) as avg_order_value
from coffee_shop_sales;
```

avg_order_value
4.686367

II Time Based Analysis

a) Monthly performance analysis

```
with monthly_performance as(
        select
            month(transaction_date) as month_num,
            monthname(transaction_date) as month_name,
            sum(transaction_qty * unit_price) as total_sales,
            count(transaction_id) as total_orders,
            sum(transaction_qty) as total_quantities_sold
        from coffee_shop_sales
        group by month(transaction_date),monthname(transaction_date)
                ),
mom_change as(
        select
            month_num,
            month_name,
            total_sales,
            lag(total_sales) over(order by month_num) as prev_month_sales,
            total_orders,
            lag(total_orders) over(order by month_num) as prev_month_orders,
            total_quantities_sold,
            lag(total_quantities_sold) over(order by month_num) as prev_month_qty
        from monthly_performance   )
select
    month_name,
    total_sales,
    round((total_sales-prev_month_sales)*100.0/nullif(prev_month_sales,0),2)
                as   mom_sales_growth,
    total_orders,
    round((total_orders-prev_month_orders)*100.0 /nullif(prev_month_orders,0),2)
                as mom_orders_growth,
    total_quantities_sold,
    round((total_quantities_sold-prev_month_qty)*100.0/nullif(prev_month_qty,0),2)
                as mom_quantities_growth
from mom_change
order by month_num;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

month_name	total_sales	mom_sales_growth	total_orders	mom_orders_growth	total_quantities_sold	mom_quantities_growth
January	81677.74	NULL	17314	NULL	24870	NULL
February	76145.19	-6.77	16359	-5.52	23550	-5.31
March	98834.68	29.80	21229	29.77	30406	29.11
April	118941.08	20.34	25335	19.34	36469	19.94
May	156727.76	31.77	33527	32.33	48233	32.26
June	166485.88	6.23	35352	5.44	50942	5.62

b) Avg Daily Sales per Month

```
with month_daily_sales as(
        select
                monthname(transaction_date) as month_name,
                day(transaction_date) as day_of_month,
                sum(transaction_qty*unit_price) as total_sales
        from coffee_shop_sales
        group by monthname(transaction_date),day(transaction_date)    )
select
        month_name,
        round(avg(total_sales),2) as daily_avg_sales
from month_daily_sales
group by month_name;
```

month_name	daily_avg_sales
January	2634.77
February	2719.47
March	3188.22
April	3964.70
May	5055.73
June	5549.53

c) Hourly sales distribution

```
select
        extract(hour from transaction_time) as hour_of_day,
        sum(transaction_qty*unit_price) as total_sales,
        sum(transaction_qty) as quantities_sold,
        count(transaction_id) as orders_placed
from coffee_shop_sales
```

```
group by hour_of_day
order by hour_of_day;
```

hour_of_day	total_sales	quantities_sold	orders_placed
6	21900.27	6865	4594
7	63526.47	19449	13428
8	82699.87	25197	17654
9	85169.53	25370	17764
10	88673.39	26713	18545
11	46319.14	14035	9766
12	40192.79	12690	8708
13	40367.45	12439	8714
14	41304.74	12907	8933
15	41733.10	12923	8979
16	41122.75	12881	9093
17	40134.31	12700	8745
18	34286.20	10826	7498
19	28446.68	8595	6092
20	2935.64	880	603

d) Trend analysis by Days of week

```
select
        dayname(transaction_date) as day_of_week,
        sum(transaction_qty*unit_price) as total_sales,
         sum(transaction_qty) as quantities_sold,
         count(transaction_id) as orders_placed
from coffee_shop_sales
group by day_of_week;
```

day_of_week	total_sales	quantities_sold	orders_placed
Sunday	98330.31	30182	21096
Monday	101677.28	31231	21643
Tuesday	99455.94	30449	21202
Wednesday	100313.54	30625	21310
Thursday	100767.78	100313.54	21654
Friday	101373.00	31207	21701
Saturday	96894.48	29614	20510

e) Sales Analysis by weekdays and weekends

```
with daytype_sales as (
    select
        case when weekday(transaction_date) between 0 and 4 then 'weekdays'
          else 'weekends'
```

```
        end as day_type,
        sum(transaction_qty*unit_price) as total_sales
    from coffee_shop_sales
    group by day_type )
select
   day_type,
   total_sales,
   round(total_sales*100.0/ sum(total_sales) over(),2) as perc_of_total
from daytype_sales;
```

day_type	total_sales	perc_of_total
weekends	195224.79	27.94
weekdays	503587.54	72.06

f) Sales Analysis by weekdays and weekends for each month

```
with sales_by_monthly_daytype as (
        select
              month(transaction_date) as month_num,
               monthname(transaction_date) as month_name,
               case when weekday(transaction_date) between 0 and 4 then 'weekdays'
                  else 'weekends'
                end as day_type,
                sum(transaction_qty*unit_price) as total_sales
        from coffee_shop_sales
        group by month_num, month_name, day_type)
select
      month_name,
       day_type,
      total_sales,
      round(total_sales*100.0/ (sum(total_sales) over(partition by month_num)),2)
              as   perc_of_monthtotal
from sales_by_monthly_daytype
order by month_num, day_type;
```

month_name	day_type	total_sales	perc_of_monthtotal
January	weekdays	58513.11	71.64
January	weekends	23164.63	28.36
February	weekdays	54002.67	70.92
February	weekends	22142.52	29.08
March	weekdays	73367.33	74.23
March	weekends	25467.35	25.77
April	weekdays	79592.51	66.92
April	weekends	39348.57	33.08
May	weekdays	116627.84	74.41
May	weekends	40099.92	25.59
June	weekdays	121484.08	72.97
June	weekends	45001.80	27.03

III Product Performance Analysis

a) Top 10 Revenue-generating Products

```
select
        product_detail,
        sum(transaction_qty*unit_price) as total_sales,
        sum(transaction_qty) as quantities_sold
from coffee_shop_sales
group by product_detail
order by total_sales desc
limit 10;
```

product_detail	total_sales	quantities_sold
Sustainably Grown Organic Lg	21151.75	4453
Dark chocolate Lg	21006.00	4668
Latte Rg	19112.25	4497
Cappuccino Lg	17641.75	4151
Morning Sunrise Chai Lg	17384.00	4346
Latte	17257.50	4602
Jamaican Coffee River Lg	16481.25	4395
Sustainably Grown Organic Rg	16233.75	4329
Cappuccino	15997.50	4266
Brazilian Lg	15109.50	4317

b) Bottom 10 products by revenue

```
select
        product_detail,
        sum(transaction_qty*unit_price) as total_sales,
```

```
        sum(transaction_qty) as quantities_sold
from coffee_shop_sales
group by product_detail
order by total_sales asc
limit 10;
```

product_detail	total_sales	quantities_sold
Dark chocolate	755.20	118
Earl Grey	1270.90	142
Spicy Eye Opener Chai	1335.90	122
Guatemalan Sustainably Grown	1340.00	134
Lemon Grass	1360.40	152
Peppermint	1369.35	153
Traditional Blend Chai	1369.35	153
English Breakfast	1440.95	161
Serenity Green Tea	1470.75	159
Morning Sunrise Chai	1596.00	168

c) Category-wise contribution to total revenue

```
with category_sales as(
        select
            product_category,
            sum(transaction_qty*unit_price) as total_sales
        from coffee_shop_sales
        group by product_category   )
select
        product_category,
        total_sales,
        round(total_sales*100.0 /(sum(total_sales) over()),2) as perc_of_total
from category_sales
order by total_sales desc;
```

product_category	total_sales	perc_of_total
Coffee	269952.45	38.63
Tea	196405.95	28.11
Bakery	82315.64	11.78
Drinking Chocolate	72416.00	10.36
Coffee beans	40085.25	5.74
Branded	13607.00	1.95
Loose Tea	11213.60	1.60
Flavours	8408.80	1.20
Packaged Chocolate	4407.64	0.63

IV Store Performance Analysis

a) Sales Analysis by Store

```
with store_sales as (
        select
                store_id,
                store_location,
                sum(transaction_qty*unit_price) as total_sales
        from coffee_shop_sales
        group by store_id, store_location  )
select
      store_id,
      store_location,
      total_sales,
      round(total_sales*100.0 /(sum(total_sales) over()),2) as perc_of_total
from store_sales
order by total_sales desc;
```

store_id	store_location	total_sales	perc_of_total
8	Hell's Kitchen	236511.17	33.84
3	Astoria	232243.91	33.23
5	Lower Manhattan	230057.25	32.92

b) Month-over-Month growth analysis at store level

```
with store_monthly_sales as (
        select
              store_id,
              store_location,
              month(transaction_date) as month_num,
              monthname(transaction_date) as month_name,
              sum(transaction_qty*unit_price) as total_sales
        from coffee_shop_sales
        group by store_id, store_location, month_num, month_name   ),
mom_change as  (
          select
                store_id,
                store_location,
                month_num,
```

```
                month_name,
                total_sales,
                lag(total_sales) over(partition by store_id order by month_num)
                    as prev_month_sales
        from store_monthly_sales   )
select
        store_id,
        store_location,
        month_name,
        total_sales,
        prev_month_sales,
        round((total_sales-prev_month_sales)*100.0/nullif(prev_month_sales,0),2)
                as mom_growth_rate
from mom_change
order by store_id, month_num ;
```

store_id	store_location	month_name	total_sales	prev_month_sales	mom_growth_rate
3	Astoria	January	27313.66	NULL	NULL
3	Astoria	February	25105.34	27313.66	-8.09
3	Astoria	March	32835.43	25105.34	30.79
3	Astoria	April	39477.61	32835.43	20.23
3	Astoria	May	52428.76	39477.61	32.81
3	Astoria	June	55083.11	52428.76	5.06
5	Lower Manhattan	January	26543.43	NULL	NULL
5	Lower Manhattan	February	25320.05	26543.43	-4.61
5	Lower Manhattan	March	32888.68	25320.05	29.89
5	Lower Manhattan	April	39159.33	32888.68	19.07
5	Lower Manhattan	May	51700.07	39159.33	32.02
5	Lower Manhattan	June	54445.69	51700.07	5.31
8	Hell's Kitchen	January	27820.65	NULL	NULL
8	Hell's Kitchen	February	25719.80	27820.65	-7.55
8	Hell's Kitchen	March	33110.57	25719.80	28.74
8	Hell's Kitchen	April	40304.14	33110.57	21.73
8	Hell's Kitchen	May	52598.93	40304.14	30.51
8	Hell's Kitchen	June	56957.08	52598.93	8.29

KEY INSIGHTS

Time-Based Demand Patterns

Monthly Trend :

- Sales exhibit clear month-over-month variation, indicating seasonality and demand fluctuation.
- May and June contribute significantly higher revenue, suggesting peak consumption periods.

Day wise Trend :

- Sales are evenly distributed across the week .
- Weekdays show higher revenue concentration than weekends.

Hourly Trend :

- Peak sales occur during morning hours i.e. 7 a.m. to 10 a.m.
- Off-peak hours show uniform transaction volume.

Product Insights

Best and lowest sellers :

- Sustainably grown organic lg, Dark Chocolate lg, Latte rg are among the best selling products
- Dark Chocolate, Earl Grey, Spicy eye opener chai are among the lowest selling products

Product Category Insight :

- Categories like Coffee, Tea, Bakery and Drinking Chocolate dominate revenue with around 80% share
- While other categories contribute less despite variety.

Store Level Performance

- Uniform distribution in performance across store locations in terms of revenue and transaction volume.
- Stores show high variation in MoM growth, suggesting operational investigation